

420-SN1-RE Programming in Science - Exercises for Topic 7 - Booleans

These are practice questions *similar* to what you *might* see on a test (some are *harder* that you will likely see on an test). These will *not* be collected or graded, but I recommend trying these as practice.

A. Given the following assignments: $p = \text{True}$, $q = \text{False}$, and $r = \text{False}$, what would be the values of the following expressions?

- | | |
|---|--|
| 1. $p \text{ or } q \text{ and } r$ | 4. $\text{not } p \text{ and } q \text{ or } r$ |
| 2. $p \text{ and } q \text{ or } r$ | 5. $p \text{ and not } q \text{ or } r$ |
| 3. $\text{not } (p \text{ and } q \text{ or } r)$ | 6. $p \text{ and } q \text{ or } r \text{ and } q$ |

B. What would each of the following Python programs print?

1.

```
x = 1
y = 1
while x < 6 or y < 100:
    y *= x
    x += 2
print(x, y)
```

2.

```
c = [ "alpha", "beta", "omicron", "pi", "delta", "lambda", "iota"]
d = []
i = 0
while i < len(c):
    v = c[i]
    n = len(v)
    if n >= 4 and n % 2 == 0:
        d = d + [v]
    i = i + 1
print(d, i)
```

3.

```
x = [5, 2, -5, 1, -9, 0, 1, 3, 8, 4, -1, 3]
total = 0
i = 0
while i < len(x):
    v = x[i]
    if abs(v) > 1 and v % 2 != 0:
        total = total + v
    i = i + 1
print(total, i, v)
```

4.

```
x = [0, 1, 5, 7, 8, 11]
y = [2, 3, 4, 6, 9]
n = len(x)
m = len(y)
i = j = k = 0 # Sets all three names to zero.
z = [0]*(n + m)
while i < n or j < m:
    if i < n and (j >= m or x[i] <= y[j]):
        z[k] = x[i]
        i = i + 1
    else:
        z[k] = y[j]
        j = j + 1
    k = k + 1
print(z)
```

C. Which of the following expressions is equivalent to the given Boolean expression? Consider using a truth table to check your answer.

1. `not a or not b`

a. `a and b`

b. `not (a and b)`

c. `not (a or b)`

d. `not a and not b`

2. `a and b or a and c`

a. `(a or b) and (a or c)`

b. `a or (b and c)`

c. `a or b or c`

d. `a and (b or c)`